

5. Information was collected about accidents on the *Seapron* bypass. It was found that the number of accidents per month could be modelled by a Poisson distribution with mean 2.5

Following some work on the bypass, the numbers of accidents during a series of 3-month periods were recorded. The data were used to test whether or not there was a change in the mean number of accidents per month.

- (a) Stating your hypotheses clearly and using a 5% level of significance, find the critical region for this test. You should state the probability in each tail. (5)
- (b) State $P(\text{Type I error})$ using this test. (1)

Data from the series of 3-month periods are recorded for 2 years.

- (c) Find the probability that at least 2 of these 3-month periods give a significant result. (3)

Given that the number of accidents per month on the bypass, after the work is completed, is actually 2.1 per month,

- (d) find $P(\text{Type II error})$ for the test in part (a) (3)



1. The number of customers entering Jeff's supermarket each morning follows a Poisson distribution.

Past information shows that customers enter at an average rate of 2 every 5 minutes.

Using this information,

- (a) (i) find the probability that exactly 26 customers enter Jeff's supermarket during a randomly selected 1-hour period one morning, (2)
- (ii) find the probability that at least 21 customers enter Jeff's supermarket during a randomly selected 1-hour period one morning. (2)

A rival supermarket is opened nearby. Following its opening, the number of customers entering Jeff's supermarket over a randomly selected 40-minute period is found to be 10

- (b) Test, at the 5% significance level, whether or not there is evidence of a decrease in the rate of customers entering Jeff's supermarket. State your hypotheses clearly. (4)

A further randomly selected 20-minute period is observed and the hypothesis test is repeated. Given that the true rate of customers entering Jeff's supermarket is now 1 every 5 minutes,

- (c) calculate the probability of a Type II error. (5)

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7. A six-sided die has sides labelled 1, 2, 3, 4, 5 and 6

The random variable S represents the score when the die is rolled.

Alicia rolls the die 45 times and the mean score, $\bar{S}$, is calculated.

Assuming the die is fair and using a suitable approximation,

- (a) find, to 3 significant figures, the value of k such that $P(\bar{S} < k) = 0.05$ (8)

- (b) Explain the relevance of the Central Limit Theorem in part (a). (2)

Alicia considers the following hypotheses:

H_0 : The die is fair

H_1 : The die is not fair

If $\bar{S} < 3.1$ or $\bar{S} > 3.9$, then H_0 will be rejected.

Given that the true distribution of S has mean 4 and variance 3

- (c) find the power of this test. (3)

- (d) Describe what would happen to the power of this test if Alicia were to increase the number of rolls of the die.
Give a reason for your answer. (2)

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5. Asha, Davinda and Jerry each have a bag containing a large number of counters, some of which are white and the rest are red.
Each person draws counters from their bag one at a time, notes the colour of the counter and returns it to their bag.

The probability of Asha getting a red counter on any one draw is 0.07

- (a) Find the probability that Asha will draw at least 3 white counters before a red counter is drawn. (2)
- (b) Find the probability that Asha gets a red counter for the second time on her 9th draw. (2)

The probability of Davinda getting a red counter on any one draw is p .
Davinda draws counters until she gets n red counters. The random variable D is the number of counters Davinda draws.

Given that the mean and the standard deviation of D are 4400 and 660 respectively,

- (c) find the value of p . (4)

Jerry believes that his bag contains a smaller proportion of red counters than Asha's bag. To test his belief, Jerry draws counters from his bag until he gets a red counter. Jerry defines the random variable J to be the number of counters drawn up to and including the first red counter.

- (d) Stating your hypotheses clearly and using a 10% level of significance, find the critical region for this test. (5)

Jerry gets a red counter for the first time on his 34th draw.

- (e) Giving a reason for your answer, state whether or not there is evidence that Jerry's bag contains a smaller proportion of red counters than Asha's bag. (2)

Given that the probability of Jerry getting a red counter on any one draw is 0.011

- (f) show that the power of the test is 0.702 to 3 significant figures. (3)



7. A manufacturer has a machine that produces lollipop sticks.

The length of a lollipop stick produced by the machine is normally distributed with unknown mean μ and standard deviation 0.2

Farhan believes that the machine is not working properly and the mean length of the lollipop sticks has decreased.

He takes a random sample of size n to test, at the 1% level of significance, the hypotheses

$$H_0: \mu = 15 \quad H_1: \mu < 15$$

- (a) Write down the size of this test.

(1)

Given that the actual value of μ is 14.9

- (b) (i) calculate the minimum value of n such that the probability of a Type II error is less than 0.05

Show your working clearly.

(6)

- (ii) Farhan uses the same sample size, n , but now carries out the test at a 5% level of significance. Without doing any further calculations, state how this would affect the probability of a Type II error.

(1)

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5. A machine fills cartons with juice.

The amount of juice in a carton is normally distributed with mean μ ml and standard deviation 8 ml.

A manager wants to test whether or not the amount of juice in the cartons, X ml, is less than 330 ml. The manager takes a random sample of 25 cartons of juice and calculates the mean amount of juice $\bar{x}$ ml.

- (a) Using a 5% level of significance, find the critical region of $\bar{X}$ for this test.
State your hypotheses clearly.

(4)

The Director is concerned about the machine filling the cartons with more than 330 ml of juice as well as less than 330 ml of juice. The Director takes a sample of 55 cartons, records the mean amount of juice $\bar{y}$ ml and uses a test with a critical region of

$$\{\bar{Y} < 328\} \cup \{\bar{Y} > 332\}$$

- (b) Find P(Type I error) for the Director's test.

(2)

When $\mu = 325$ ml

- (c) find P(Type II error) for the test in part (a)

(2)

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2. The number of errors made by a secretary is modelled by a Poisson distribution with a mean of 2.4 per 100 words.

A 100-word piece of work completed by the secretary is selected at random.

- (a) Find the probability that

(i) there are exactly 3 errors,

(ii) there are fewer than 2 errors.

(2)

After a long holiday, a randomly selected piece of work containing 250 words completed by the secretary is examined to see if the rate of errors has changed.

- (b) Stating your hypotheses clearly, and using a 5% level of significance, find the critical region for a suitable test.

(4)

- (c) Find $P(\text{Type I error})$ for the test in part (b)

(1)

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7. Sam and Tessa are testing a spinner to see if the probability, p , of it landing on red is less than $\frac{1}{5}$. They both use a 10% significance level.

Sam decides to spin the spinner 20 times and record the number of times it lands on red.

- (a) Find the critical region for Sam's test. (2)

- (b) Write down the size of Sam's test. (1)

Tessa decides to spin the spinner until it lands on red and she records the number of spins.

- (c) Find the critical region for Tessa's test. (6)

- (d) Find the size of Tessa's test. (1)

- (e) (i) Show that the power function for Sam's test is given by

$$(1 - p)^{19}(1 + 19p)$$

- (ii) Find the power function for Tessa's test. (4)

- (f) With reference to parts (b), (d) and (e), state, giving your reasons, whether you would recommend Sam's test or Tessa's test when $p = 0.15$ (4)